📦 Solving Inventory Inefficiencies Using Advanced SQL Analytics

Urban Retail Co. – A Data-Driven Inventory Optimization Project

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🚀 Introduction

Urban Retail Co. is a mid-sized but rapidly growing retail chain operating across both physical outlets and e-commerce platforms. With over **5,000 unique SKUs** spanning groceries, personal care, electronics, and home essentials, the company relies on a distributed warehousing and store network to fulfill customer demand across various regions.

Despite this scale, Urban Retail is currently plagued by inventory inefficiencies due to **manual processes** and **underutilized data**. This project aims to **unlock the potential of structured SQL analytics** to optimize stock levels, reduce losses, and drive operational excellence.

🧩 The Challenge

Key issues faced by Urban Retail include:

- ❌ Frequent stockouts of fast-moving products, leading to missed sales and customer dissatisfaction
- 📦 Overstocking of slow-moving items, increasing storage cost and tying up working capital
- 🕶️ Lack of real-time visibility into inventory trends and supplier performance
- 📉 Reactive (not predictive) inventory decisions across stores and categories

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🎯 Objective

Our goal is to **build a scalable, SQL-driven analytics pipeline** that extracts insights from raw transactional and warehouse data to help:

- Detect inventory blind spots
- Automate reorder calculations
- Highlight inefficiencies across time, regions, and SKUs
- Generate KPIs to guide smarter, faster, data-driven decisions

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🛠️ Solution Approach

We built a complete **ETL + analytics solution** using MySQL, split across three architectural layers:

🔷 1. Bronze Layer – Raw Data Ingestion

- **Database:** `inventory_bronze_layer`
- **Table:** `raw_inventory_data`
- **Script:** `bronze_ddl.sql`
- **Data Load:** `proc_load_bronze.sql` using `LOAD DATA INFILE`

This layer captures raw CSVs and prepares them for transformation, ensuring ingestion scalability.

🔶 2. Silver Layer – Normalized Relational Schema

- **Database:** `inventory_silver_layer`
- **Schema Setup:** `silver_ddl.sql`
- **Data Transformation & Load:** `proc_load_silver.sql`

Structured into normalized tables:

- `Products`, `Stores`, `Inventory`, `Sales`, `Orders`, `Weather`, `Forecasts`
- Designed with **referential integrity**, **foreign keys**, and **composite keys**.

🔎 Data Quality Checks

Before populating the Silver Layer, we perform validation logic:

- ✅ Deduplication using `ROW_NUMBER()` in CTEs
- ✅ Trimming and standardizing string fields
- ✅ Ensuring unique `Product ID`, `Store ID` + `Region` pairs
- ✅ Eliminating repeated `Weather` records by `(store, region, date)`
- ✅ Validating date consistency across datasets

These steps ensure accurate joins, KPIs, and consistent downstream analytics.

📊 3. Analytical Layer – Business-Critical Insights

Stored in modular `.sql` scripts inside the `analysis/` folder:

✅ Stock Level Summary (`stock_summary.sql`)

- Total, regional, and store-level inventory rollups
- Identifies understocked and overstocked products

⚠️ Low Inventory Detection (`low_inventory.sql`)

- Calculates 7-day reorder point using 30-day sales history
- Flags products below threshold with latest stock info

📦 Reorder Point Estimation (`reorder_point.sql`)

- `ROP = Average Daily Demand × 7` (with optional safety buffer)
- Empowers predictive purchasing

🔄 Inventory Turnover Analysis (`turnover_ratio.sql`)

- Turnover ratio per product-store-month
- Ratings: High / Moderate / Low turnover

📈 KPI Summary Report (`kpi_summary.sql`)

- Stockout rate (% of days with zero stock)
- Inventory age (duration tracked)
- Average inventory levels across time

All queries use modern SQL features like **CTEs**, **aggregations**, and **window functions**.

📂 Project Structure

```
project/
│
├── bronze_ddl.sql
├── proc_load_bronze.sql
│
├── silver_ddl.sql
├── proc_load_silver.sql
│
├── analysis/
│   ├── stock_summary.sql
│   ├── low_inventory.sql
│   ├── reorder_point.sql
│   ├── turnover_ratio.sql
│   └── kpi_summary.sql
│
└── summary
```

📊 Output Impact

By implementing this analytics-driven system, Urban Retail gains:

- ⚠️ **Real-time low stock alerts** across stores
- 🔀 **Smart reorder recommendations** based on demand patterns
- 🏷️ **Categorical insights** into fast-moving vs. slow-moving SKUs
- 🕰️ **Inventory aging reports** for smarter clearance decisions
- 💡 **KPI-driven dashboards** to support leadership and supply chain planning
- 💰 Reduced holding costs and improved working capital utilization
- 📦 Enhanced stock availability → better customer satisfaction

This pipeline lays the foundation for **BI tools** like **Power BI, Metabase, or Looker** to visualize and automate inventory health monitoring.

🎓 Submitted To

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Project Tag: `@caciitg`
Prepared as part of the Summer Projects 2025 initiative.

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ThankYou!!